THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. For the data with classes 200–220, 220–240, 240–260, 260–280, 280–300, 300–320, 320–340 with frequencies 23, 4, 25, 6, 3, 19, 20 respectively, The mean of the data is: 1

(a) 270 **(b)** 269.8 **(c)** 270.8 **(d)** 268.8

2. A bag contains 15 red balls and x blue balls. The probability of drawing a blue ball is 3 times the probability of drawing a red ball. The value of x is: 1

(a) 3 **(b)** 44 **(c)** 60 **(d)** 45

3. The smallest positive integer by which 12 must be multiplied to obtain a perfect square is: 1

(a) 2 **(b)** 6 **(c)** 12 **(d)** 3

4. If α, β are zeroes of $x^2 - 6x + 8$, then $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ is: 1

(a) 20 **(b)** $\frac{9}{2}$ **(c)** $\frac{5}{2}$ **(d)** $\frac{3}{4}$

5. The area (in sq. units) of the triangle formed by the line $2x + 3y = 12$ and the coordinate axes is: 1

(a) 72 **(b)** 12 **(c)** 3 **(d)** 24

6. If $x = 1$ is a root of $kx^2 + 3x - 5 = 0$, then $k =$ 1

(a) 2 **(b)** 5 **(c)** -2 **(d)** -8

7. The sum of the first n terms of an AP is $S_n = 2n^2 + 3n$. Its common difference is: 1

(a) 4 **(b)** 3 **(c)** 5 **(d)** 2

8. If $\triangle ABC \sim \triangle PQR$, $ar(ABC) = 18$ and $ar(PQR) = 98$ and $AB = 6$, then $PQ =$ 1

(a) 6 **(b)** 17 **(c)** 98 **(d)** 14

9. The ratio in which $(-3, 2)$ and $(6, 5)$ are joined by the y-axis is 3: 1

(a) 3 **(b)** 6 **(c)** 5 **(d)** 9

10. If $\sec\theta + \tan\theta = 3$, then $\sec\theta - \tan\theta =$ 1

(a) 1 **(b)** $\frac{1}{3}$ **(c)** 3 **(d)** 2

11. From the top of a tower $6\sqrt{3}$ m high, the angles of depression of two objects on level ground, on the same side of the tower and in line with its foot, are 30° and 60°. The distance between the two objects is: 1

(a) 6 m **(b)** 12 m **(c)** $12\sqrt{3}$ m **(d)** 24 m

12. Two concentric circles have radii 68 cm and 32 cm. A chord of the larger circle touches the smaller circle. The length of the chord is: 1

(a) 60 cm **(b)** 120 cm **(c)** 72 cm **(d)** 100 cm

13. A sector of a circle of radius 42 cm has central angle $45°$. The perimeter of the sector is: 1

(a) 75 cm **(b)** 117 cm **(c)** 159 cm **(d)** 33 cm

14. A solid cone and a solid hemisphere have equal base radii 7 cm and equal volumes. The height of the cone is: 1

(a) 7 cm **(b)** 10.5 cm **(c)** 21 cm **(d)** 14 cm

15. For the data with classes 20–40, 40–60, 60–80, 80–100, 100–120, 120–140, 140–160 with frequencies 3, 13, 11, 17, 15, 24, 17 respectively, The median of the data is: 1

(a) 120 **(b)** 100 **(c)** 103.6 **(d)** 108

16. A bag contains 7 red and 9 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{4}{7}$. The value of x is: 1

(a) 4 **(b)** 6 **(c)** 10 **(d)** 5

17. Which of these is a rational number? 1

(a) $\sqrt{4}$ **(b)** $2\sqrt{5}$ **(c)** $\sqrt{2}+\sqrt{3}$ **(d)** $\frac{\sqrt{12}}{\sqrt{3}}$

18. For the cubic with zeroes $1, 2, 3$, $\alpha\beta+\beta\gamma+\gamma\alpha$ is: 1

(a) -11 **(b)** 6 **(c)** 11 **(d)** 36

19. If $x+y=7$ and $xy=10$, then $x^2+y^2=$ 1

(a) 49 **(b)** 69 **(c)** 39 **(d)** 29

20. The equation $kx^2+6x+3=0$ has real roots when: 1

(a) $k<3$ only **(b)** $k>0$

(c) $k\leq 3$ and $k\neq 0$ **(d)** $k\geq 3$

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. The mean of a distribution with classes 200–220, 220–240, 240–260, 260–280, 280–300, 300–320, 320–340 and frequencies 3, 24, p, 10, 15, 14, 17 respectively is 274. Find the value of p. 2

22. Two fair dice are thrown together and the numbers on the upper faces are noted. Find the probability that the sum is a multiple of 4. 2

23. Find the smallest positive integer that leaves remainders 7 and 13 when divided by 12 and 18 respectively. 2

24. If the sum of the zeroes of x^2-6x+k is equal to their product, find k. 2

25. Solve: $5x+3y=47$ and $3x+5y=41$. 2

26. If α, β are the roots of $x^2-6x+5=0$, find $\alpha^2+\beta^2$. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Find a and b for which the pair $3x + y = 7$ and $(a+b)x + (a-b)y = 14$ has infinitely many solutions. 3

28. For what values of k does $x^2 + 5x + k = 0$ have real roots? Find the greatest integer value of k. 3

29. Find the sum of the first 16 terms of the AP whose 3rd term is 14 and 8th term is 34. 3

30. Find the altitude of an equilateral triangle of side 6 cm, in terms of $\sqrt{3}$. 3

31. The line joining $(0,\,0)$ and $(9,\,6)$ is trisected by points P and Q. Find their coordinates. 3

OR

If $\sin\theta + \cos\theta = \frac{7}{5}$, find $\sin\theta\cos\theta$. 3

32. A flagstaff stands on the top of a building 14 m high. From a point on level ground the angles of elevation of the bottom and the top of the flagstaff are 45° and 60° respectively. Find the length of the flagstaff. 3

OR

In $\triangle ABC$, right-angled at B, $AB = 36$ cm and $BC = 77$ cm. A circle is inscribed in the triangle. Find the radius of the inscribed circle. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. A loan is repaid in 18 monthly instalments; the first is ₹2500 and each later instalment is ₹150 more than the previous. Find the instalment in the last month and the total amount repaid. 5

34. Two poles of heights 30 m and 12 m stand on level ground 24 m apart. Find the distance between their tops. 5

35. Three vertices of a parallelogram $ABCD$ taken in order are $A(-2,\,1)$, $B(1,\,0)$ and $C(4,\,3)$. Find the fourth vertex D. 5

36. If $\tan A = \frac{24}{7}$, evaluate $\dfrac{5\sin A + 4\cos A}{7\sin A + 3\cos A}$. 5

OR

Two vertical poles of equal height stand on opposite sides of a road 48 m wide. From a point on the road between them, the angles of elevation of the tops of the poles are 60° and 30°. Find the common height of the poles and the distances of the point from the poles (nearer pole first). 5

37. Tangents PA and PB are drawn from a point P, 75 cm from the centre O of a circle of radius 72 cm. The length of each tangent is 21 cm. Find the length of the chord AB. 5

OR

Two concentric circles have radii 21 cm and 14 cm. A sector of angle 36° is marked in both. Find the area of the region between the two arcs within the sector. 5

38. A solid sphere of radius 31.5 cm fits exactly inside a cylinder (it touches the curved surface and both bases). Find the volume of the space in the cylinder not occupied by the sphere. 5

OR

The marks obtained by students in a test are given below. 5

Marks	40–50	50–60	60–70	70–80	80–90	90–100	100–110	110–120
Number of students	7	23	19	2	9	16	13	11

Find the mean and the mode (in this order).

END OF QUESTION PAPER **Total: 80 Marks**

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